

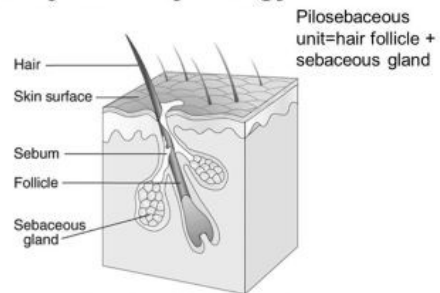
Sebaceous Skin Disorders

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IS-GERD

Learning Objectives

- After this lecture the medical student should be able to:
 - Acne vulgaris and some of its variants
 - Rosacea
 - Perioral dermatitis
 - Hidradenitis Suppurativa/Acne inversa
- Recognize the description, or a picture, of a typical clinical presentation
- Describe the most commonly affected areas
- Identify the key epidemiological features that differentiate these skin disorders
- Describe the pathogenesis if known
- List the appropriate therapeutic agents and strategies

Anatomy and Physiology of skin



Pilosebaceous Unit: most commonly on face, chest and back

Pilosebaceous Unit



Case Presentation #1

- A 15 yr old male, otherwise healthy, presents to your office for “skin problems” for the last 6 months.
- He reports pimples on his forehead and cheeks, that sometimes have pus. He also notices them on his back. He tried a facewash he bought at the pharmacy, but it is not helping.
- What do you suspect his diagnosis is? [acne vulgaris](#)



Acne Vulgaris

Vulgaris means “common”

Epidemiology: Who does acne affect?

- Affects up to 95% of adolescents in westernized societies. 20% will have severe cases.
- Mild/moderate: female > male. Severe cases: males > females.
- Acne may be seen in newborns.
- Acne can continue into middle age. 12% Women, 3% Men
- A family history of severe or nodular acne may increase risk of developing acne.

Newborns develop acne due to the exposure to estrogen/androgen now thought to be due to immature adrenal glands and the production of higher estrogen/androgen levels. In the past, classic theories thought neonatal acne was due to high exposure to estrogen during pregnancy that resolved weeks-months after birth.

Acne scars



Pitted scars due to acne vulgaris are present on the cheek and mandible.

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Clinical Significance

- Psychological impact: cosmetic, social interaction, self-confidence.
- Embarrassment: Teens are not likely to bring it up or seek healthcare for it.
- Importance to support patient through the process
- Scarring is long term- recognize and treat early! **prevent long term sequelae**

Acne Vulgaris

- Manifests as skin lesions called comedones and progressing into papulopustules, nodules.
- Erythema is present if there is inflammation.
- Appears in certain body areas (face, trunk, rarely buttocks)
- Can result in pitted, depressed or hypertrophic scars



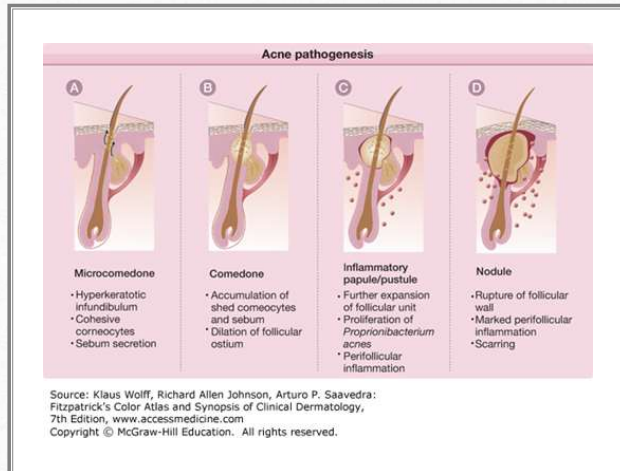
The upper back of males with acne vulgaris frequently presents pustular and cystic lesions.



Small, inflamed papules and pustules and hyperpigmented macules on the cheek.

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Pathogenesis of Acne

- This is COMPLEX relevant



❖ The Follicular orifice becomes blocked by keratinous material and sebum (become microcomedones then comedones) The formation of comedones occurs.

◆ Androgen-stimulated increased sebum production and proliferation of the microorganism *Cutibacterium acnes* (*C. acnes*, formerly *Propionibacterium acnes*), an anaerobic diphtheroid that is a normal component of skin flora **occur within the follicular plug.**

◆ *Cutibacterium acnes* acts to release free fatty acids from sebum, an inflammation reaction ensues.

◆ Causes sterile inflammation within the comedones and

- ◆ results in rupture of the nodule wall
- ◆ Further inflammatory reaction results as oily and keratinous debris from the nodule are extruded into the dermis.

Formation of a Comedone (non-inflammatory)



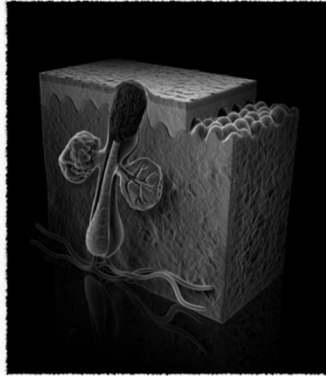
- Hyper production of sebum in the sebaceous gland
- Formation of a plug (keratinization) blocking the pilosebaceous unit



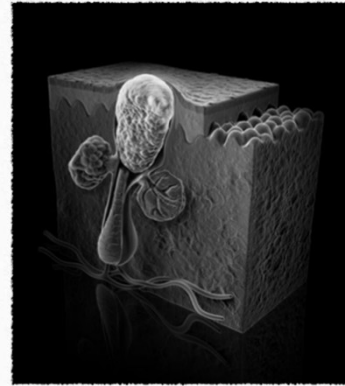
board relevant

Comedones (non-inflammatory)

Open= "Blackhead"
Dilated pore, oxidized oily debris



Closed= "Whitehead"
1-2mm, pebbly white papule



Mild acne



This 14-year-old girl has multiple closed comedones and scattered, small, inflammatory papules and pustules on the forehead.

Courtesy of Douglas Hoffman, MD, Dermatlas;
<http://www.dermatlas.org>.

Comedonal acne



Numerous open comedones are present on the chin.

Reproduced with permission: Goodheart HP. Goodheart's Photoguide of Common Skin Disorders, 2nd ed, Lippincott Williams & Wilkins, Philadelphia 2003. Copyright © 2003 Lippincott Williams & Wilkins.

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Note the difference again between closed comedones and open comedones.

Development into Inflammatory Acne



Inflammation, erythema is visible on exam.

Inflammatory reaction to *Cutibacterium acnes*
anaerobic bacterium and component of skin flora.

C. Acnes acts to release free fatty acids from sebum

Causes sterile inflammation within the comedones
and results in rupture of the nodule wall

Follicular wall rupture and spread of perifollicular
inflammation

Acne Vulgaris Characteristics



- Lesions: Multiple types- comedones, papules, pustules (scabs); if severe: nodules, sinus tracts
- Color: pink, red, brown/black.
- Margination: Individual lesions are well-defined
- Darker skin tones: hyperpigmentation resulting from inflammation more likely
- Consistency: Some firm, some fluctuant
- Shape: round
- Arrangement: Multiple scattered discrete lesion
- Distribution of lesions: Regional, pilosebaceous units, (face, trunk)

Acne Differential Diagnosis



- Face
 - Pseudofolliculitis barbae
 - Rosacea
 - Perioral dermatitis
- Trunk
 - Malassezia folliculitis
 - “Hot tub” pseudomonas folliculitis
 - Staph aureus folliculitis
 - Acne-like conditions

Everything on this list will create red lesions, consider if they are raised, associated with a pilosebaceous unit or not. For Acne, it is the presence of comedones and the common locations.

Acne Myths & Truths

- MYTH: Certain foods can worsen acne. Contrary to common belief, studies have shown no relation to diet (e.g. chocolate, fatty food, alcohol)
- Insulin resistance may stimulate androgen production which may increase sebum.
- MYTH: acne is not caused by dirt or bad hygiene. In fact, excessive washing can make it worse (breaks down the natural skin oils)
- TRUTH: Stress can make acne worse.
- TRUTH: Occlusion/pressure like leaning on one's hands can exacerbate acne. i.e. "maskne"

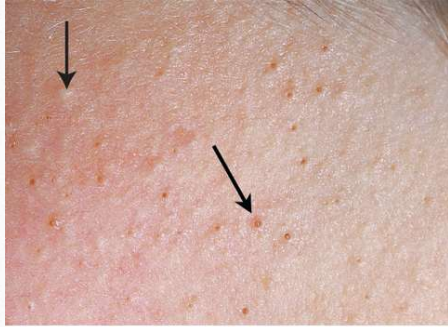


Hyperinsulinemia effect may play a role in the development of acne- Study of Japanese women in Japan vs living in US, western population diet with more acne

Acne Classification

- **Noninflammatory**
 - Open and closed comedones
- **Inflammatory**
 - Presence of one or more of the following: papules, pustules, and nodules (cysts)
 - Classified as papulopustular and/or nodular
 - Severity: mild, moderate, severe

Acne Classification



Source: Klaus Wolff, Richard Allen Johnson, Arturo P. Saavedra:
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology,
7th Edition, www.accessmedicine.com
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Noninflammatory
Comedones



Source: Klaus Wolff, Richard Allen Johnson, Arturo P. Saavedra:
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology,
7th Edition, www.accessmedicine.com
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Inflammatory
Papulopustular



usually very tender and erythematous with swelling

Acne Conglobata

- Severe form of nodular acne
- Uncommon, but ask about family history
- Nodules (may be very painful), sinus tracts
- May result in severe scarring

probably need specialist



Follicular occlusion tetrad is a symptom complex consisting of four conditions having a similar pathophysiology. It includes Hidradenitis suppurativa, acne conglobata, dissecting cellulitis of the scalp and pilonidal sinus.

if see one may want to check others

Acne Diagnosis

- Diagnosis is clinical, comedones +/- inflammation and pustules, nodules on face or back.
- Labs/imaging are usually unnecessary unless you suspect an underlying systemic reason like PCOS, hyperandrogenism (other clinical signs would be present)



Source: S.C. Taylor, A.P. Kelly, H.W. Lim, A.M.A. Serrano
Taylor and Kelly's Dermatology for Skin of Color, Second Edition
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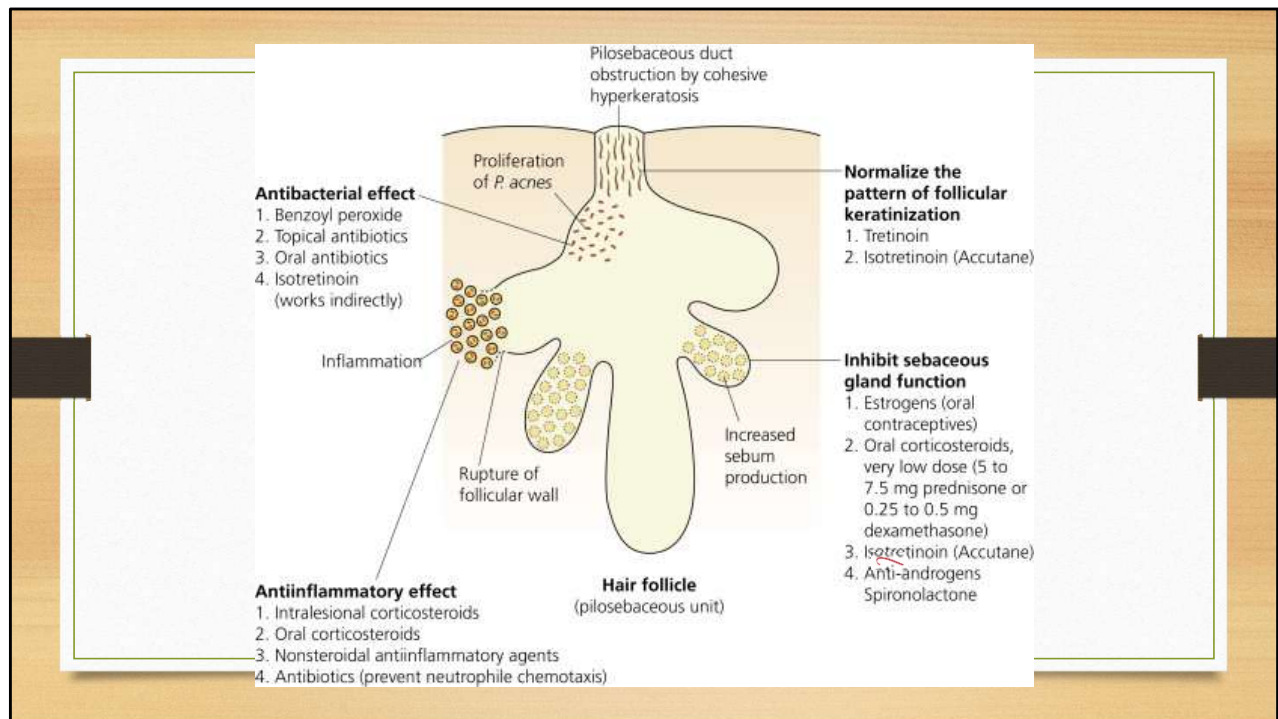
A close-up photograph of a person's eye and cheek. A glass dropper is shown applying a clear liquid to the cheek. The background is a soft, out-of-focus skin tone.

Treatment Recommendations

- Review your patient's whole skin routine, educate about use of gentle cleaners and other topicals is important
- Show proper application for topical therapy
- Help patients with realistic expectations of therapy, it takes time to see results!
- Empathy for patient distress, support.

Acne Treatment

- **Goals of treatment:**
 - Remove follicular plugs
 - Reduce sebum production
 - Treat bacterial colonization
- **Medication strategy:** (COMBO treatment is helpful)
 - Start with topical treatments first (retinoids, benzoyl peroxide)
 - Topical antibiotics
 - Oral antibiotics
 - Oral contraceptives (hormonal acne)
 - Isotretinoin (reserved for nodulocystic acne)



Treatment of closed comedones is slow and they may remain unchanged for months. Benzoyl peroxide is thought to reduce bacterial resistance.

start topically - retinoid, benzoyl peroxide then to oral then hormonal then to isotretinoin

Acne Treatment

- **Topical**

usually otc but everything else rx
 Benzoyl peroxide gels or washes (antibacterial effect), Azelaic Acid

- Topical antibiotics: clindamycin or erythromycin
- Topical Retinoids: tretinoin, adapalene

- Oral Antibiotics: Minocycline or Doxycycline

- Hormonal manipulation: females: OCPs or Spironolactone

- Intralesional triamcinolone: for inflammatory cysts and nodules

- Oral retinoids: isotretinoin (Accutane)

bleach towels so tell them to wash hands

steroid injections if really painful but usually defer to derm

Azelaic acid is used as a topical treatment for mild to moderate acne, and may be combined with oral antibiotics or hormonal therapy. It is useful for both comedonal acne and inflammatory acne.

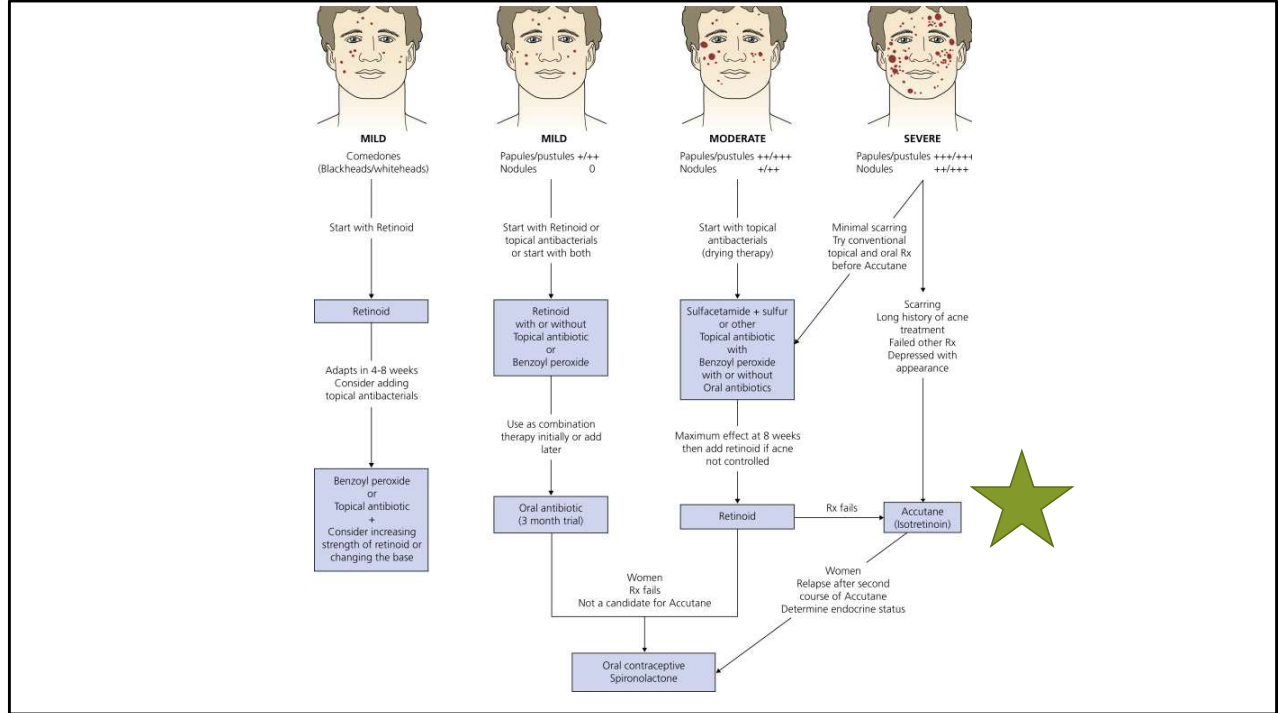
Azelaic acid is:

Antibacterial - it reduces the growth of bacteria in the follicle (*C. acnes* and *Staphylococcus epidermidis*)

Keratolytic and comedolytic - it returns to normal the disordered growth of the skin cells lining the follicle

A scavenger of free radicals - i.e. it reduces inflammation.

Azelaic acid also helps reduce pigmentation, so it's particularly useful for darker skinned patients whose acne spots leave persistent brown marks (postinflammatory pigmentation) or who have melasma.



don't need to memorize

Mild Acne Treatment

- ◆ Start with a retinoid
- ◆ Increase strength if irritation does not occur
- ◆ Add benzoyl peroxide or topical antibiotics or combo meds to discourage *C. acnes* and the formation of inflammatory lesions
- ◆ Treatment of closed comedones is slow and they may remain unchanged for months



MILD
Comedones
(Blackheads/Whiteheads)

Start with Retinoid

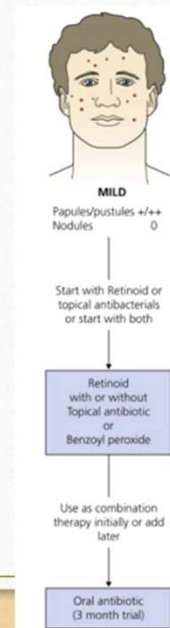
Retinoid

Adapts in 4-8 weeks
Consider adding
topical antibacterials

Benzoyl peroxide
or
Topical antibiotic
+
Consider increasing
strength of retinoid or
changing the base

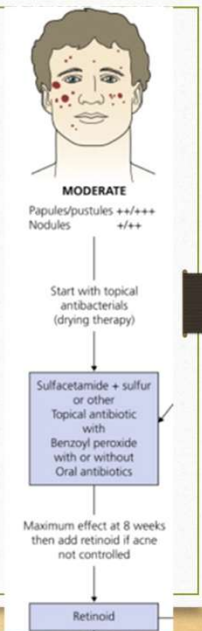
Mild Inflammatory Acne

- Start with benzoyl peroxide, a topical antibiotic or combo medicine AND a retinoid applied on alternate evenings
- Use low concentrations initially
- After the adjustment period, use a retinoid each night and benzoyl peroxide or antibiotic each morning
- Oral Abx can be used if the number of pustules does not decrease



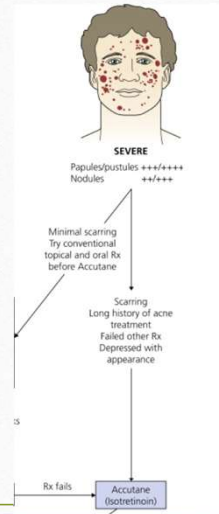
Moderate-to-Severe Inflammatory Acne

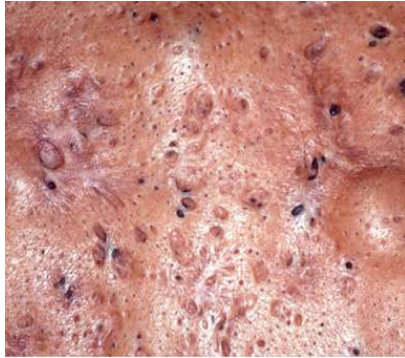
- ◆ Begin with a topical retinoid and a combo with a topical antibiotic
- ◆ Oral Abx can be used in patients with more than 10 pustules
- ◆ Treatment should be continued until no new lesions develop (2-4 mo) then slowly tapered



Severe Nodulocystic Acne

- ◆ Topical retinoids or antibiotic
- ◆ Systemic therapy with oral antibiotics or isotretinoin





Source: S.C. Taylor, A.P. Kelly, H.W. Lim, A.M.A. Serrano
for and Kelly's Dermatology for Skin of Color, Second Edition
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Isotretinoin (Accutane)

- Highly controlled regimen- 20 week course usually with dermatologist.
- Prolonged remissions (1-3 yrs) are seen in about 80% of cases
- Side effects: dry skin, chapped lips, dry eyes, nosebleeds, hair loss, hyperlipidemia, **Depression**
- **Category X! Teratogenic. Females need to be on birth control** 2 types and sign off
- [iPLEDGE program](#)

verify pregnancy test and 2 forms of birth control

Acne Pearls

- ◆ Always ask your patients what they have tried and what the outcome was
- ◆ Common side effects:
 - ◆ Dry skin: creams Oily skin: gels or sometimes higher concentrations
 - ◆ Benzoyl Peroxide: stains clothes and bedding
 - ◆ Azelaic acid- may help with hyperpigmentation
 - ◆ Oral antibiotics: doxycycline – sun sensitivity, tetracyclines can yellow teeth (consider in women of reproductive age)
 - ◆ Isotretinoin- category X teratogen.



Case Presentation #1

- A 15 yr old male, otherwise healthy, presents to your office for “skin problems” for the last 6 months.
- He reports pimples on his forehead and cheeks, that sometimes have pus. He also notices them on his back. He tried some facewash he bought at the pharmacy, but it is not helping.
- On exam you see multiple papules with erythematous base and some open comedones.
- What do you suspect his diagnosis is?

Mild inflammatory acne- BP wash, topical retinoin, topical antibacterial combo

Case Presentation #2

- A 45 yr old female with no past medical history presents to your office to address red raised lesions on her cheeks. She admits to having acne as a teenager. She is worried that her acne has returned.
- She reports that drinking alcohol and eating certain foods seem to make the redness on her face worse. She tried using some medicated face wash with no improvement.





Rosacea

Rosacea

Chronic
inflammatory
disorder of
pilosebaceous units

Etiology is
unknown

Affects about 10%
of fair skinned
persons

Onset typically 30
to 50 yrs old.

Clinical finding of
rhinophymas more
common in men

Mostly affects
women

bypass pre-teen

Once thought to be related to acne but not related.

Theory that immune response is perpetuated, from some genetic factors, calthecidin levels are higher with individuals with roseacea. May be more prevalent as some of the flushing features are less appreciated on darker skinned individuals, more research needed.

Rosacea



Inflammatory papules and pustules are present on the nose and cheeks.

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Rosacea

- Episodes of increased capillary reactivity with increase in skin temperature
- Rosacea diathesis: "Flushing and blushing"
- Triggers: hot liquids in mouth, spicy foods, alcohol, sun or external heat exposure.
- Appearance often leads to an assumption that patient has alcoholic dependence- not true.

Rosacea Characteristics

Type of skin: tiny papules, papulopustules, **no comedones or nodules**

Color: erythematous lesions on erythematous background

Margination: Well-defined lesions on ill-defined background

Consistency: papules are slightly firm

Shape: papules are generally round, “dome shaped”

Arrangement: multiple, regular grouping, some confluent

Distribution of lesions: **localized to face** (cheeks, nose, chin, forehead)

Rosacea Classification

- ◆ Erythematotelangiectatic rosacea (ETR)
- ◆ Papulopustular rosacea
- ◆ Phymatous rosacea
- ◆ Ocular rosacea

Erythematotelangiectatic (ETR)

- Chronic redness of the nose and medial cheeks
- Flushing
 - May be associated with sweating or transient facial swelling
- Telangiectasia's may be visible and blanch
- Rough and scaling skin may be present
- Increased skin sensitivity and triggers



Telangiectasia: small blood vessels seen at the surface of the skin.

Papulopustular Rosacea

- Lesions may be mistaken for acne vulgaris
- **Unlike acne, no comedones occur**
- Lesions confined to face



commonly see in women
no surrounding erythema associated with le:



Phymatous Rosacea

Rhinophyma

Chronic active inflammation causes tissue hypertrophy and distortion manifesting as thickened skin with irregular contours

Can be seen on the chin (gnathophyma), forehead (glabellophyma) and cheeks

Majority are adult men

<https://www.thieme-connect.com/products/ejournals/abstract/10.1055/s-2008-1064455>



Wolff K, Goldsmith LA, Katz SI, Gilchrist BA, Paller AS, Leffell DJ:

Ocular Rosacea

- Ocular involvement occurs in more than 50% of patients with rosacea
- May proceed, follow or occur concurrently with cutaneous disease
- Conjunctival hyperemia, anterior blepharitis, keratitis, lid margin telangiectasias, abnormal tearing, cicatricial conjunctivitis, and chalazion or hordeolum (stye) formation
- Refer to ophthalmologist **special wash, ophtho preventing further injury**

Rosacea Differential Diagnosis

Facial papules/pustules

- Acne
- Perioral dermatitis
- Folliculitis (staph aureus or gram-negative)
- Demodex folliculorum (mite) infestation

Facial flushing/erythema

- Seborrheic dermatitis
- Prolonged use of topical glucocorticoids
- Systemic lupus erythematosus
- Dermatomyositis

Acne Vulgaris vs. Rosacea



OTHER FEATURES: Acne Vulgaris

- Most prevalent in adolescents and young adults
- Variable distribution on face
- Frequent shoulder, chest, and/or back involvement
- Sequelae of postinflammatory hyperpigmentation, postinflammatory erythema, and scarring
- Association with hyperandrogenic disorders (eg, polycystic ovarian syndrome)

OTHER FEATURES: Rosacea

- Most prevalent in adults >30 years old
- Centrofacial distribution (cheeks, nose, chin)
- Ocular involvement (eg, symptoms of eye irritation, eyelid erythema, conjunctival injection, crusting, recurrent hordeolum or chalazion)
- Sensitive skin
- Flushing

KEY CONCEPTS

Acne vulgaris and rosacea are common causes of inflamed papules or pustules on the face. Recognition of other characteristic features is helpful for distinguishing these conditions. Patients may exhibit some or all of the displayed features.

Distinguishing between acne vulgaris and rosacea is important because of differences in the approach to patient evaluation and treatment. For example, an assessment for signs of associated hyperandrogenism (eg, menstrual irregularity, hirsutism, virilization) is an important component of the initial evaluation of female patients with acne vulgaris, particularly in the presence of severe, sudden-onset, or recalcitrant acne. In patients with rosacea, an assessment for signs or symptoms of ocular involvement is important for identifying patients who may benefit from ophthalmologic examination.

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Rosacea Assessment

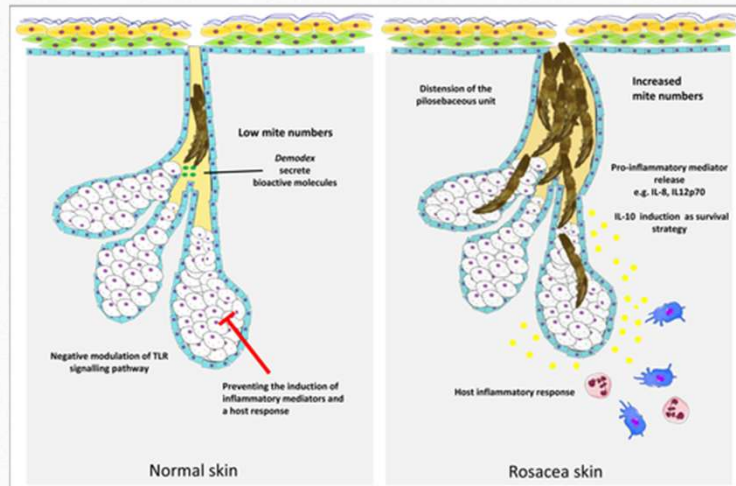
Diagnosis is clinical

Biopsy may be needed to r/o
other disorders

Culture for bacteria if clinically
indicated

Skin scraping for Demodex
folliculorum

Demodex mite infection



British Journal of Dermatology 2018

When treatment is refractory, skin scrapings may show large numbers of Demodex mites. Patients may respond to ivermectin 1% cream or a single dose of oral ivermectin

rosacea that's not responding

Rosacea Treatment

- **Topical (mild cases)**
 - Metronidazole gel or cream
 - Sodium sulfacetamide
 - Azelaic acid
 - Ivermectin cream (demodex folliculorum)
- **Oral (papules or ocular involvement)**
 - Doxycycline or Minocycline
 - Metronidazole
 - Ivermectin
 - Isotretinoin (category X) reserved for severe cases
 - Carvedilol has been reported to reduce erythema and telangiectasia
- **Procedures**
 - Moderate to Severe disease
 - Vascular lasers
 - Intense pulsed light
 - **Phymatous sub-type**
 - Cold scalpel tangential excision
 - Heated scalpel excision
 - Electrocautery
 - Dermabrasion
 - Laser ablation
 - Radio frequency electrosurgery

Steroids will make rosacea worse! Never apply a topical steroid to the rosacea as although short-term improvement may be observed (vasoconstriction and anti-inflammatory effect), it makes the rosacea more severe over the next weeks (possibly by increased production of nitric oxide)- Dermnet NZ.

Rosacea Patient Education

- Expect recurrences over a few years but disease may disappear spontaneously
- Use gentle skin products and emollients
- Use sunscreen
- Identify and avoid triggers
 - Sunlight
 - Stress
 - Particular foods or beverages



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Perioral Dermatitis

Perioral Dermatitis

Epidemiology

- Affects mostly women in 20-30s
- If seen in children, males and females affected equally
- Occasionally seen in older adults

Idiopathic!

- Asthma inhalers/topical steroids
- Gum, toothpastes, diet

Symptoms

- Asymptomatic or itching
- Persists for months
- Can develop after long term use of moderate or strong topical steroids

Perioral Dermatitis

- Type of skin lesion: tiny papules, papulovesicles, and papulopustules. **No comedones!**
- Color: reddish on an erythematous base
- Margination: discrete lesions
- Consistency: slightly firm
- Shape: papules are round
- Arrangement: irregularly grouped, some confluent.
- Distribution: **Localized to perioral area and nasolabial folds, chin or lateral ocular canthi**
- **Spares a narrow zone around vermillion border**
- Mild scaling is possible



Perioral Dermatitis DDX



Acne vulgaris



Rosacea



Allergic contact dermatitis



Irritant contact dermatitis



Lupus miliaris disseminatus faciei



Perioral Dermatitis

- **Hallmark is exacerbation when treated with potent corticosteroids**
- Misdiagnosis is common and steroid treatment is common for other skin disorders in differential
- Flare may occur after steroids are discontinued (within 1 week) and then start to have with gradual improvement



Perioral Dermatitis

Clinical presentation can range from mild to severe

Perioral Dermatitis Treatment

- First step: Stop any topical glucocorticoids!
- Topical- control and maintenance.
 - Metronidazole or erythromycin gel
 - Calcineurin inhibitors
- Systemic/Oral antibiotics
 - Minocycline or Doxycycline
 - Tetracycline
 - Use high dose until clear then reduced dose for 2 months.

Beware as tetracycline side effect is yellowing of teeth in children, doxycycline, sun sensitivity.

May recur once
treatment is
stopped but will
resolve again
with treatment

Avoid any
topical
corticosteroids
(including OTC
hydrocortisone)

Avoid heavy
facial creams

Use mild soap
or soap
substitute

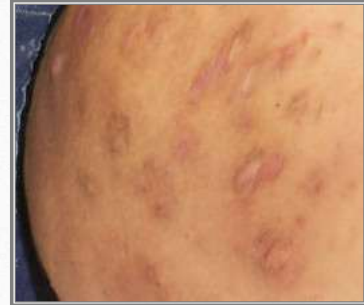
Perioral Dermatitis Patient Education



Hidradenitis Suppurativa

Hidradenitis Suppurativa

- Inflammatory skin disease that affects pilosebaceous skin in the axillae, in the groin, perianal region, and under the breasts.
- It is characterized by recurrent boil-like nodules and abscesses that culminate in pus-like discharge, difficult-to-heal open wounds (sinuses) and scarring.



- ◆ Chronic suppurative (forming/discharging pus) cicatrices (scarring) disease of apocrine gland-bearing skin
- ◆ Axillae, genitofemoral, perianal regions (rarely the scalp)
- ◆ More accurately a disorder of the follicle

- ◆ Not solely an infectious disorder
- ◆ May be associated with severe nodular acne

Hidradenitis Suppurativa

- Prevalence in US is 1-2%
- Starts at puberty and most active in the 20-40 year range.
- Females more likely to be affected than males, 3:1 ratio.
- Up to a third of patients have a family history of HS
- Hidradenitis suppurativa also has a significant psychological impact, and many patients suffer from impairment of body image, depression and anxiety.

often missed in primary care



Association not causation

folliculitis in axilla, peri-anal often go to ER not PCP and get tx but recurrence keeps happening and can I scarring

make sure to ask about and examine!!

Hidradenitis Suppurativa DDX

Early:

- Furuncle (involves entire pilosebaceous unit)
- Carbuncle (multiple units: confluence of furuncles)
- Lymphadenitis
- Ruptured inclusion cyst
- Painful lymphadenopathy in lymphogranuloma venereum or cat-scratch disease

Late:

- Lymphogranuloma venereum
- Granuloma Inguinale
- Scrofuloderma
- Actinomycosis
- Sinus tracts and fistulas associated with ulcerative colitis and regional enteritis



Hurley Staging of Hidradenitis Suppurativa

Figure 1 Hurley Stages of Lesions in Hidradenitis Suppurativa. Although the examples shown here are of hidradenitis suppurativa of the axillary area, the same principles of the Hurley staging system apply to the disease when it affects other locations.

Panel A shows stage I disease, which is localized and includes the formation of single or multiple abscesses, without sinus tracts and scarring.

Panel B shows stage II disease, which is characterized by recurrent abscesses, with sinus tract formation and scarring, occurring as either single lesions or multiple, widely separated lesions.

Panel C shows stage III disease, which includes diffuse or nearly diffuse involvement of the affected region, with multiple interconnected tracts and abscesses across the entire area.



Figure 2 Involvement of the Genitalia in a Woman with Hurley Stage III Disease. Genitofemoral lesions are common in patients with hidradenitis suppurativa, especially in women.

very painful



Involvement of the penis, scrotum, perineum and thighs in a patient with Hurley stage III disease



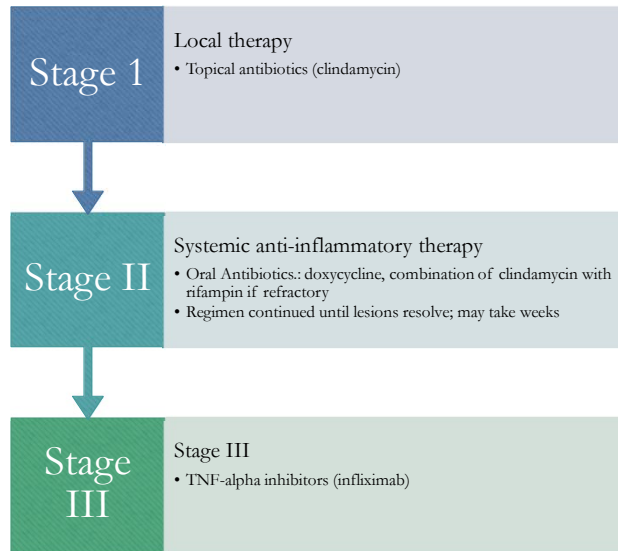
Figure 3 Involvement of the Buttocks in a Man with Hurley Stage III Disease. The disease may also involve the buttocks, in addition to the axillae and genitofemoral folds, which are the pathognomonic locations.

Hidradenitis Suppurativa: Assessment

- General approach
 - Treatment and prognosis depend on the stage
 - Psychological consequences may be very severe so aggressive plan and reassurance are needed
- Diagnosis is clinical
 - May need culture material from refractory lesions to optimize antibiotic choice

Interdisciplinary approach is helpful

Hidradenitis Suppurativa: Treatment



Hidradenitis Suppurativa: Treatment

- All stages may benefit from prednisone if pain and inflammation are severe (pulse with high dose for 2-3 days, then tapered over 2 weeks)
- Oral retinoids
 - Not useful in severe disease
 - May be useful in early disease to prevent follicular plugging (especially in multimodal approach with surgical excision of individual lesions)

Hidradenitis Suppurativa: Treatment

For acute painful lesions:

- Intralesional corticosteroid injection (triamcinolone)
- With incision & drainage if abscess

Surgical management

- Excise chronic recurrent, fibrotic nodules or sinus tracts
- Excision of lesions with skin grafting

Patient education

- Psychological burden- addressing depression/anxiety
- Smoking cessation- may see improvement in 6 months.

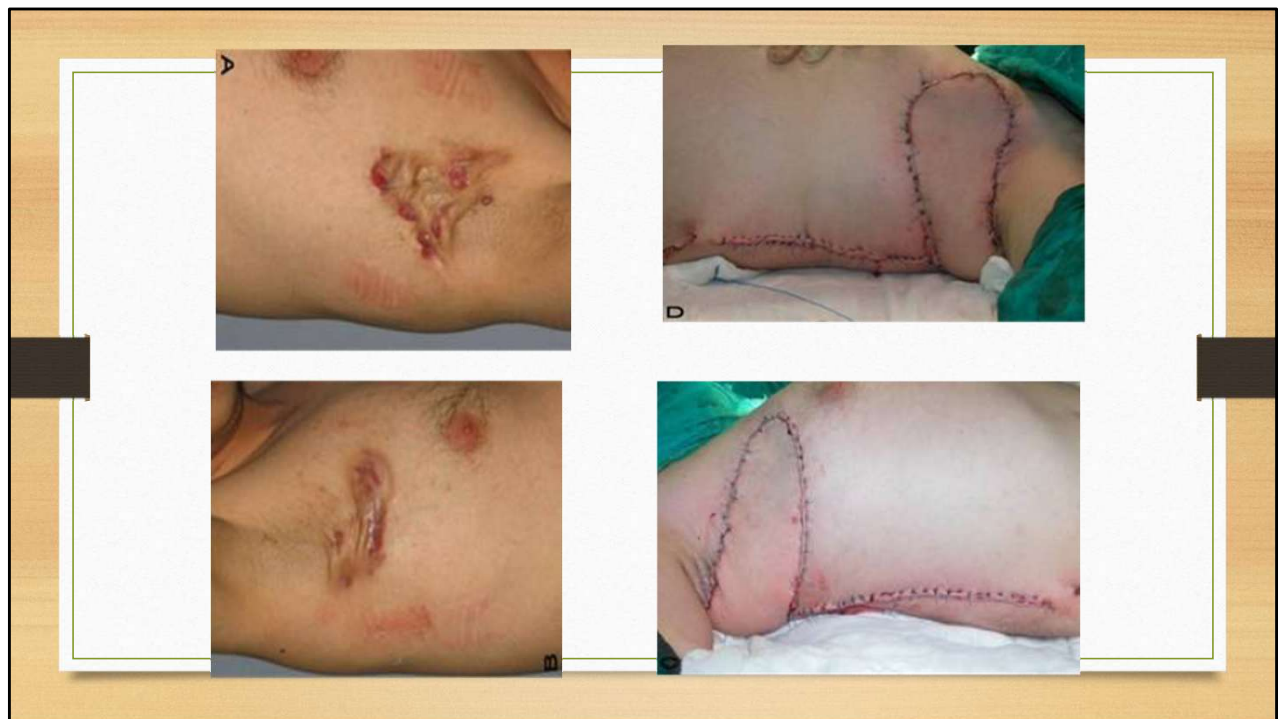
I & D should not be routine treatment

Tylenol for pain can help.

Rambhatla et al⁵⁰ put forth a suggested evidence-based treatment approach based on systematic review of the literature regarding treatment options for HS. For Hurley stage I disease, topical clindamycin 1% and treatment of lesions with laser are recommended therapeutic options.

For stage

II disease, a 10-week course of combination clindamycin and rifampin or three to four monthly treatments with laser are suggested. If the disease progresses despite these therapies, or if the patient is unable to tolerate these treatments, escalation to use of a biologic medication is suggested, with infliximab being the treatment of choice. For Hurley stage III disease, a trial of the same treatment modalities mentioned for stage II disease is recommended. In the case of failure or unsatisfactory response, it is recommended that the patient be referred to a surgeon for evaluation for wide local excision of the affected area.⁵⁰ Ultimately, the therapeutic ladder should be reviewed on a case-by-case basis by the physician with each patient and the risks and benefits associated with each treatment discussed thoroughly before deciding on a treatment plan.



before and after surgery with skin grafts

early dx important, getting to specialist

References

- Basics: Fitzpatrick's Color Atlas & Synopsis of Clinical Dermatology, 7e (AccessMedicine)
Part 1, Section 1: Disorders of Sebaceous and Apocrine Glands
- Details: Fitzpatrick's Dermatology in General Medicine, 8e (AccessMedicine)
Part 3, Section 13. Disorders of the Sebaceous Glands
Part 3, Section 14. Disorders of the Eccrine and Apocrine Glands
- DermIS <http://www.dermis.net/dermisroot/en/home/index.htm>
- Acne vulgaris (updated 1/23/15) <http://emedicine.medscape.com/article/1069804-overview>
- Acneiform eruptions (updated 2/28/14) <http://emedicine.medscape.com/article/1072536-overview>
- Rosacea (updated 8/21/14) <http://emedicine.medscape.com/article/1071429-overview>
- Perioral dermatitis (updated 7/11/14) <http://emedicine.medscape.com/article/1071128-overview>
- Hidradenitis suppurativa, Dermnet NZ <https://dermnetnz.org/topics/hidradenitis-suppurativa/>
- Jemec JBE. Hidradenitis suppurativa. Clinical Practice. New Engl J Med 2012;366:158-64. <http://www.nejm.org/doi/full/10.1056/NEJMcp1014163>
- Menderes A, Sunay O, Vayvada H, Yilmaz M. Surgical Management of Hidradenitis Suppurativa. *Int J Med Sci* 2010; 7(4):240-247. Available from <http://www.medsci.org/v07p0240.htm>
- Habif TP. *Clinical Dermatology*, 5th ed. Elsevier, 2009. (Available online via Touro's electronic library link with ClinicalKey)
- UpToDate.com



Case 1

- A 45 year old male presents to your office for acne. He states he had problems with acne when he was a teenager but recently it came back. He states that for the last couple months, he has had red spots on his nose and cheeks that get worse when he is in the sun or drinks alcohol. He has tried OTC face washes and acne treatments but it has not helped.
- On exam, you see multiple, tiny, firm, red papules on an erythematous background on cheeks, nose, and extending to the forehead. There are no comedones present.
- Which of the following is the most likely diagnosis?
 - Acne Vulgaris
 - Acne Conglobata
 - Rosacea
 - Contact dermatitis

Case 1 con't

- After discussing your diagnosis with your patient. You recommend a topical medication.
- Which medication would you prescribe?
 - A) benzoyl peroxide
 - B) triamcinolone ointment
 - C) metronidazole gel
 - D) isotretinoin

What do you recommend regarding the triamcinolone ointment?

Metronidazole gel is the first choice for roseacea. Steroids may provide initial benefit but then worsen rosacea, avoid steroids!

Case 2

◆ A 16-year old girl returns to your clinic with skin findings similar to that seen in Figure. She reports a trial of Retin A and topical antibiotics with little improvement. You are considering treatment with oral medication.

◆ What test must this patient have prior to starting oral antibiotics?

- a) Complete blood count
- b) Pregnancy test
- c) Chest x-ray
- d) Basic metabolic panel
- e) No testing needed



Moderate acne with pustules. You would consider moving to oral antibiotics like doxycycline or minocycline.

It is recommended to check w urine pregnancy test before prescribing.

Case 3



- A 25 year old female presents to your office with “skin infection”. She states she has developed several hard bumps in her bilateral armpits. One has opened and started draining fluid. She has stopped using deodorant because she thinks it may be making them worse. She states the lesions are very painful and are making it hard to move her arms comfortably.
- On exam, you find several hard nodules in her bilateral axilla. There is one pustule in the left axilla that is open and draining white, viscous fluid.
- Which of the following is the next step in her care?
 - Discontinue all deodorant
 - Oral antibiotic therapy
 - Use of topical benzoyl peroxide
 - Referral to plastic surgery

Stage II, sinus tracts, multiple oral antibiotics would be next step

Case 4



- A 29 year old female presents to your office with rash. She states for the last several weeks she has had extremely dry skin around her mouth and chin. She has a history of eczema so she tried to use some OTC hydrocortisone ointment on her face that she had left over. This made it worse.
- On exam, you see tiny, firm, round, red papules on an erythematous background, irregularly grouped around her perioral area and nasolabial folds.
- Which of the following is true about her condition?
 - It should be treated with benzoyl peroxide
 - It should be treated with high potency topical steroids
 - Steroids should be avoided
 - It is due to underlying acne vulgaris

Perioral dermatitis, avoid steroids.